

A Unified Kalman-Type Classification of Controllability for Linear Stochastic Systems

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Abstract

We provide a unified algebraic classification of approximate controllability, approximate null controllability, exact null controllability, and exact controllability for finite-dimensional linear stochastic control systems with constant coefficients and multidimensional Brownian noise. The controls may enter both the drift and the diffusion terms, and the joint diffusion-control matrix may have arbitrary rank. A finite decreasing sequence of subspaces defines a common geometric obstruction. Its vanishing characterizes the two approximate properties and exact null controllability. Exact controllability to arbitrary random targets additionally requires the joint diffusion-control matrix to be surjective. The codimension of the obstruction is given by a rank difference of explicit polynomial block matrices, yielding finite tests in the original coefficients. When the joint diffusion-control matrix is surjective, we obtain a finite Kalman-type full-rank condition for exact controllability. If the obstruction vanishes, a linear state feedback with a deterministic time-dependent gain drives every initial state to zero at any prescribed positive time with finite energy and polynomial small-time cost. An equivalent orthogonal elimination algorithm computes the rank differences with polynomial storage and cost, avoiding exponentially large block matrices. A two-noise example relates the input classification to terminal errors and control energy in penalized stochastic linear-quadratic control problems.

Keywords. Stochastic systems; Controllability; Multiplicative noise; State feedback; Rank conditions; Numerical algorithms

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1 Introduction

Linear stochastic systems with state- and control-dependent diffusion provide a tractable setting for studying feedback design in the presence of multiplicative uncertainty. Such models arise, for example, when actuation signals are transmitted through uncertain input channels [18]. For multi-input systems, the organization and allocation of these channels also enter the control design, as illustrated by scheduling/control co-design over shared stochastic channels [6]. This model also connects mean-square stabilization with generalized Riccati equations [27]. These connections motivate a structural question for finite-horizon control: which input configurations permit a prescribed terminal requirement to be met by an adapted control with finite energy? In particular, reaching the origin at a prescribed time and reaching an arbitrary random terminal state impose different requirements on how the control acts on the drift and the diffusion.

1.1 Formulation of the controllability problems

Let $W = (W^1, \dots, W^m)$ be an m -dimensional Brownian motion on a complete probability space $(\Omega, \mathcal{F}, \mathbb{P})$, and let $\{\mathcal{F}_t\}_{t \geq 0}$ be the usual augmentation of its natural filtration. Write $\mathbb{F}$ for the progressive σ -algebra and $\mathbb{E}$ for expectation.

For a Euclidean space H , let $L^2_{\mathcal{F}_t}(\Omega; H)$ denote the square-integrable $\mathcal{F}_t$ -measurable random variables. For $T > 0$, the space $L^2_{\mathbb{F}}(0, T; H)$ consists of progressively measurable processes with $\mathbb{E} \int_0^T |v(t)|_H^2 dt < \infty$. We use $L^2_{\mathbb{F}}(\Omega; C([0, T]; H))$ for continuous adapted processes with $\mathbb{E} \sup_{0 \leq t \leq T} |v(t)|_H^2 < \infty$.

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For $t \in [0, T]$, consider the linear control system

$$\begin{cases} dy(t) = [Ay(t) + Bu(t)] dt + \sum_{j=1}^m [C_j y(t) + D_j u(t)] dW^j(t), \\ y(0) = y_0, \end{cases} \quad (1.1)$$

where $A, C_j \in \mathbb{R}^{n \times n}$, $B, D_j \in \mathbb{R}^{n \times r}$, $n, r, m \in \mathbb{N}$, $j = 1, \dots, m$. The control $u \in \mathcal{U}_T := L^2_{\mathbb{F}}(0, T; \mathbb{R}^r)$.

By [16, Theorem 3.2], for every $y_0 \in \mathbb{R}^n$ and $u \in \mathcal{U}_T$, the control system (1.1) has a unique solution $y(\cdot; y_0, u) \in L^2_{\mathbb{F}}(\Omega; C([0, T]; \mathbb{R}^n))$ satisfying

$$\mathbb{E} \sup_{0 \leq t \leq T} |y(t; y_0, u)|^2 \leq c(T) \left(|y_0|^2 + \mathbb{E} \int_0^T |u(t)|^2 dt \right). \quad (1.2)$$

Let us recall the following definition.

Definition 1.1. For $T > 0$, system (1.1) is said to be:

- (EC) *exactly controllable* if, for every $y_0 \in \mathbb{R}^n$ and $\xi \in L^2_{\mathcal{F}_T}(\Omega; \mathbb{R}^n)$, there is $u \in \mathcal{U}_T$ such that $y(T; y_0, u) = \xi$, $\mathbb{P}$ -a.s.;
- (AC) *approximately controllable* if, for every $y_0 \in \mathbb{R}^n$, $\xi \in L^2_{\mathcal{F}_T}(\Omega; \mathbb{R}^n)$, and $\varepsilon > 0$, there is $u \in \mathcal{U}_T$ such that

$$\mathbb{E} |y(T; y_0, u) - \xi|^2 < \varepsilon;$$

- (ENC) *exactly null controllable* if, for every $y_0 \in \mathbb{R}^n$, there is $u \in \mathcal{U}_T$ such that $y(T; y_0, u) = 0$, $\mathbb{P}$ -a.s.;
- (ANC) *approximately null controllable* if, for every $y_0 \in \mathbb{R}^n$ and $\varepsilon > 0$, there is $u \in \mathcal{U}_T$ such that

$$\mathbb{E} |y(T; y_0, u)|^2 < \varepsilon.$$

Write $\mathcal{C} = [C_1^\top \ \dots \ C_m^\top]^\top \in \mathbb{R}^{mn \times n}$ and define the joint diffusion-control matrix

$$\mathcal{D} = [D_1^\top \ \dots \ D_m^\top]^\top \in \mathbb{R}^{mn \times r}. \quad (1.3)$$

For deterministic linear systems with constant coefficients, the Kalman rank condition characterizes controllability, and exact, null and approximate controllability coincide when the controls and targets are deterministic.

The stochastic counterpart of this classical picture is less complete. Even for linear stochastic differential equations with constant coefficients and controls entering both the drift and diffusion, the deterministic equivalences and rank condition do not carry over directly. For the class of systems in (1.1), two basic questions concern the relationships among the controllability notions and their algebraic characterization.

The first concerns the relationships among the four properties in Definition 1.1. For scalar Brownian noise, Goreac [11, Theorem 3 and Remark 4] characterized approximate controllability and approximate null controllability by a common conditional-invariance obstruction. In contrast, approximate controllability does not imply exact controllability to arbitrary random targets, as illustrated by Example 1.1. The question addressed here is whether approximate null controllability also guarantees exact null control with finite energy at every prescribed positive time for (1.1). This gap is not merely technical. The terminal control map takes values in an infinite-dimensional space of random variables. Approximate null controllability places the negative of the uncontrolled terminal state in the closure of the range of the terminal control map, whereas exact null controllability requires membership in the range itself. A sequence of approximately null controls need not have uniformly bounded energy, so the passage to exact null controllability is not automatic even in finite dimensions. We also ask what additional condition is needed to attain arbitrary random targets exactly.

The second question concerns a Kalman-type rank characterization in the original coefficients. For scalar noise and a diffusion-control matrix of full row rank, Peng [19, pp. 281–282 and Theorem 3.2] obtained a necessary and sufficient rank condition involving products of coefficient matrices after an equivalent change of controls. Lü and Zhang [16, Theorem 6.10] proved that products of length at most $n - 1$ suffice, and raised the problem of a rank characterization directly for the original system in [16, Remark 6.11]. We seek a

unified finite rank characterization in the original coefficients for multidimensional noise and arbitrary rank of $\mathcal{D}$, keeping the same adapted L^2 control class.

We resolve these two questions by identifying a common geometric obstruction and expressing its codimension as the rank difference of explicit polynomial block matrices in $A, B, (C_j, D_j)_{j=1}^m$.

From a design perspective, a criterion in the original coefficients can be applied to candidate input configurations before constructing a controller. Our results distinguish the feasibility of returning the state exactly to the origin at a prescribed time from that of attaining arbitrary square-integrable random terminal states. They also reveal a concrete restriction on the input structure: every controllability property considered here requires $B \ker \mathcal{D} \neq \{0\}$, so some input combination must affect the drift without adding diffusion. Exact controllability further requires $\text{rank } \mathcal{D} = mn$, and consequently $r \geq mn + 1$; see Remark 3.2. The complete criteria in Corollary 1.5 can thus test feasibility before a minimum-energy steering problem is solved. The numerical example in subsection 4.3 illustrates these distinctions through terminal-error and control-energy comparisons.

1.2 Main results of this paper

We first state the geometric classification and the feedback result, then express the classification through finite rank conditions in the original coefficient matrices.

We use $\langle \cdot, \cdot \rangle$ and $|\cdot|$ for Euclidean inner products and norms, and $\|\cdot\|$ and $\|\cdot\|_F$ for matrix operator and Frobenius norms. The standard coordinate vectors are denoted by $e_1, \dots, e_n$. The transpose is denoted by $\top$, and $\otimes$ denotes the Kronecker product. For a subspace V , let $V^\perp$ be its Euclidean orthogonal complement and Π_V its orthogonal projector. Matrices with zero rows or columns are allowed, with the usual rank, range and kernel conventions. In what follows, c denotes a positive constant depending only on $A, B, (C_j, D_j)_{j=1}^m$, and $c(T)$ may additionally depend on T ; these generic constants may change between formulas. Constants whose identity is used subsequently are named separately.

To describe the obstruction to controllability, for $\eta \in L^2_{\mathcal{F}_T}(\Omega; \mathbb{R}^n)$, consider the adjoint BSDE

$$\begin{cases} dz(t) = -\left(A^\top z(t) + \sum_{j=1}^m C_j^\top Z_j(t)\right) dt + \sum_{j=1}^m Z_j(t) dW^j(t), \\ z(T) = \eta. \end{cases} \quad (1.4)$$

Here $Z = (Z_1, \dots, Z_m)$. By [16, Theorem 4.2], (1.4) has a unique solution with $z \in L^2_{\mathbb{R}}(\Omega; C([0, T]; \mathbb{R}^n))$ and $Z \in L^2_{\mathbb{R}}(0, T; (\mathbb{R}^n)^m)$. Applying Itô's formula to $\langle y, z \rangle$ and taking expectations gives the duality identity

$$\mathbb{E}\langle y(T), \eta \rangle - \langle y_0, z(0) \rangle = \mathbb{E} \int_0^T \left\langle u, B^\top z + \sum_{j=1}^m D_j^\top Z_j \right\rangle dt. \quad (1.5)$$

Thus $B^\top z + \sum_{j=1}^m D_j^\top Z_j$ is the adjoint observation associated with the control. An adjoint solution with zero observation is undetected by every admissible control in (1.5).

For such an adjoint motion to remain in a subspace $V \subset \mathbb{R}^n$, each martingale coefficient w_j and the drift must belong to V . We therefore define $\mathcal{R}(V)$ as the set of $v \in V$ for which some $w_1, \dots, w_m \in V$ satisfy

$$B^\top v + \sum_{j=1}^m D_j^\top w_j = 0, \quad A^\top v + \sum_{j=1}^m C_j^\top w_j \in V. \quad (1.6)$$

Starting from $\mathcal{V}_0 = \mathbb{R}^n$, set

$$\mathcal{V}_{k+1} = \mathcal{R}(\mathcal{V}_k) \quad (k \geq 0). \quad (1.7)$$

We denote $\mathcal{V}_n$ by $\mathcal{V}_*$.

The next lemma gives an intrinsic characterization of the limiting space.

Table 1: The only possible joint regimes under the standing assumptions.

$\mathcal{V}_*$	$\text{rank } \mathcal{D}$	Properties
$\neq \{0\}$	arbitrary	none
$\{0\}$	$< mn$	AC, ANC, ENC
$\{0\}$	mn	AC, ANC, ENC, EC

Lemma 1.1. *The sequence $(\mathcal{V}_k)_{k \geq 0}$ decreases and is stationary from step n onward. Its limit $\mathcal{V}_*$ is the largest subspace V for which some $\Gamma_1, \dots, \Gamma_m \in \mathbb{R}^{n \times n}$ satisfy*

$$\Gamma_j V \subset V \quad (j = 1, \dots, m), \quad \left(A^\top + \sum_{j=1}^m C_j^\top \Gamma_j \right) V \subset V, \quad \left(B^\top + \sum_{j=1}^m D_j^\top \Gamma_j \right) V = \{0\}. \quad (1.8)$$

Proof. For every subspace V , the linear conditions in (1.6) make $\mathcal{R}(V)$ a subspace of V . Thus $\mathcal{V}_{k+1} \subset \mathcal{V}_k$ for every $k \geq 0$. If equality holds at some step, then the sequence is stationary from that step onward. Otherwise, the dimension strictly decreases at each step; since the ambient space has dimension n , stationarity must occur no later than step n .

Next we identify the limit $\mathcal{V}_*$. At this fixed point, for each basis vector $v \in \mathcal{V}_*$, choose $w_1, \dots, w_m$ satisfying (1.6). Since the defining condition is linear, extending these assignments linearly on $\mathcal{V}_*$ and then to $\mathbb{R}^n$ defines matrices $\Gamma_1, \dots, \Gamma_m$ satisfying (1.8) with $V = \mathcal{V}_*$.

Conversely, suppose that a subspace V satisfies (1.8) for some $\Gamma_1, \dots, \Gamma_m$. We show by induction on k that $V \subset \mathcal{V}_k$. For $k = 0$, this holds since $\mathcal{V}_0$ is the ambient space. Assume $V \subset \mathcal{V}_k$. For $v \in V$, set $w_j = \Gamma_j v$. Then the invariance conditions ensure that $v \in \mathcal{V}_{k+1}$, because the defining condition for $\mathcal{V}_{k+1}$ is satisfied with these w_j . Hence $V \subset \mathcal{V}_{k+1}$. Therefore $V \subset \mathcal{V}_k$ for all k , and thus $V \subset \mathcal{V}_*$. This proves that $\mathcal{V}_*$ is the largest subspace with the stated property. $\square$

Each step removes directions that fail the zero-observation and drift conditions in (1.6). At the limit, (1.8) provides linear choices $w_j = \Gamma_j v$ in the obstruction space.

The following theorem classifies the four controllability properties through the obstruction space and the joint diffusion rank.

Theorem 1.2. *For system (1.1), the following statements hold.*

- (i) *For every $T > 0$, approximate controllability, approximate null controllability, and exact null controllability are equivalent, and each holds if and only if $\mathcal{V}_* = \{0\}$.*
- (ii) *For every $T > 0$, exact controllability to arbitrary targets in $L^2_{\mathcal{F}_T}(\Omega; \mathbb{R}^n)$ holds if and only if*

$$\mathcal{V}_* = \{0\} \quad \text{and} \quad \text{rank } \mathcal{D} = mn. \quad (1.9)$$

In particular, each of the four controllability properties is independent of the positive time horizon.

The next theorem realizes exact null controllability by a deterministic linear feedback at each prescribed terminal time and gives a polynomial small-time energy bound.

Theorem 1.3. *Assume that $\mathcal{V}_* = \{0\}$. Then for every $T > 0$ there exists a deterministic, piecewise-constant matrix function $K_T : [0, T) \rightarrow \mathbb{R}^{r \times n}$, locally bounded on $[0, T)$, such that, for every $y_0 \in \mathbb{R}^n$, the feedback control $u(t) = K_T(t)y(t)$ belongs to $\mathcal{U}_T$ and its state satisfies $y(T; y_0, u) = 0$, $\mathbb{P}$ -a.s. Moreover, there exist $c_0 > 0$ and an integer $N \geq 1$, depending only on $A, B, (C_j, D_j)_{j=1}^m$, such that*

$$\mathbb{E} \int_0^T |u(t)|^2 dt \leq c_0 T^{-N} |y_0|^2, \quad 0 < T \leq 1. \quad (1.10)$$

The gain K_T may become unbounded as $t \uparrow T$, with switching times accumulating at T . The energy bound concerns the control $u = K_T y$.

The following example illustrates the role of the joint diffusion rank in Table 1.

Example 1.1. Take $n = 1$, $r = 3$, $m = 2$, $A = C_1 = C_2 = 0$, and $B = (1, 0, 0)$, $D_1 = (0, 1, 0)$, $D_2 = (0, 1, \varepsilon)$, where $\varepsilon \in \mathbb{R}$. Then (1.1) becomes

$$dy = u_1 dt + u_2 dW^1 + (u_2 + \varepsilon u_3) dW^2.$$

The zero-observation constraint is

$$B^\top v + D_1^\top w_1 + D_2^\top w_2 = \begin{pmatrix} v \\ w_1 + w_2 \\ \varepsilon w_2 \end{pmatrix} = 0.$$

Thus $\mathcal{V}_1 = \{0\}$ for every ε . By Theorem 1.2, AC, ANC and ENC hold at every $T > 0$, whereas EC holds if and only if $\varepsilon \neq 0$: the rank of $\mathcal{D}$ is one at $\varepsilon = 0$ and two otherwise. Both D_1 and D_2 , as well as their horizontal concatenation, have full row rank even at $\varepsilon = 0$.

We next express the geometric classification through finite rank tests in the given coefficient matrices, and obtain a Kalman-type full-rank criterion for exact controllability when $\mathcal{D}$ is surjective.

Recall the definition of $\mathcal{D}$ in (1.3). Set

$$\mathbf{K}_1 = B^\top, \quad \mathbf{N}_1 = \mathcal{D}^\top. \quad (1.11)$$

For $k \geq 1$, write $\mathbf{K}_k \in \mathbb{R}^{a_k \times n}$ and $\mathbf{N}_k \in \mathbb{R}^{a_k \times b_k}$, and form

$$\mathbf{K}_{k+1} = \begin{pmatrix} B^\top \\ 0_{ma_k \times n} \\ \mathbf{K}_k A^\top \end{pmatrix}, \quad \mathbf{N}_{k+1} = \begin{pmatrix} \mathcal{D}^\top & 0 & 0 \\ I_m \otimes \mathbf{K}_k & I_m \otimes \mathbf{N}_k & 0 \\ \mathbf{K}_k \mathcal{C}^\top & 0 & \mathbf{N}_k \end{pmatrix}, \quad (1.12)$$

where $a_k = r \sum_{\ell=0}^{k-1} (m+1)^\ell$ and $b_k = mn \sum_{\ell=0}^{k-1} (m+1)^\ell$. The three column blocks of $\mathbf{N}_{k+1}$ have widths mn , mb_k and b_k , and its three row blocks have heights r , ma_k and a_k . Put $\delta_0 = 0$ and, for $k \geq 1$,

$$\delta_k = \text{rank}[\mathbf{N}_k \ \mathbf{K}_k] - \text{rank} \mathbf{N}_k. \quad (1.13)$$

The block pair represents each obstruction space and determines its codimension.

Theorem 1.4. For every $k \geq 1$,

$$\mathcal{V}_k = \{v \in \mathbb{R}^n : \exists \alpha \in \mathbb{R}^{b_k}, \quad \mathbf{K}_k v + \mathbf{N}_k \alpha = 0\}, \quad (1.14)$$

and

$$\delta_k = n - \dim \mathcal{V}_k. \quad (1.15)$$

At step n , this identity yields the controllability criterion.

Corollary 1.5. For every $T > 0$, approximate controllability, approximate null controllability, and exact null controllability are each equivalent to

$$\text{rank}[\mathbf{N}_n \ \mathbf{K}_n] - \text{rank} \mathbf{N}_n = n. \quad (1.16)$$

Exact controllability is equivalent to (1.16) together with $\text{rank} \mathcal{D} = mn$.

Proof. Apply Theorems 1.2 and 1.4 with $\mathcal{V}_* = \mathcal{V}_n$. □

When $\text{rank} \mathcal{D} = mn$, let $J = \mathcal{D}^\top (\mathcal{D} \mathcal{D}^\top)^{-1}$ and partition $J = [J_1 \ \cdots \ J_m]$, where $J_j \in \mathbb{R}^{r \times n}$. Define $E_j = B J_j$, $F = A - B J \mathcal{C}$, and $H = B(I_r - J \mathcal{D})$. Set $H_0 = H$ and, for $k \geq 0$, define

$$H_{k+1} = [E_1 H_k \ \cdots \ E_m H_k \ F H_k]. \quad (1.17)$$

The matrix H_k has n rows and $r(m+1)^k$ columns, corresponding to all products of exactly k factors from $\{E_1, \dots, E_m, F\}$ followed by H . Thus the concatenation of $H_0, \dots, H_{n-1}$ has $r \sum_{k=0}^{n-1} (m+1)^k$ columns.

These matrices give a Kalman-type rank condition for exact controllability.

Theorem 1.6. Assume that $\text{rank} \mathcal{D} = mn$. At every $T > 0$, system (1.1) is exactly controllable if and only if

$$\text{rank}[H_0 \ H_1 \ \cdots \ H_{n-1}] = n. \quad (1.18)$$

1.3 Relation to previous work

The spaces in (1.6) and (1.7) extend Goreac’s finite subspace iteration for AC and ANC [11, Theorem 3 and Remark 4] from scalar to multidimensional Brownian noise, allowing arbitrary rank of $\mathcal{D}$. Theorem 1.2 further shows that the vanishing of the same obstruction characterizes ENC at every prescribed $T > 0$.

Guo, Yu and Zhang [13] characterize stabilizability through approximate null controllability with cost: for each prescribed contraction factor δ , the control horizon may be chosen as $T = T(\delta)$. In Theorem 1.3, a linear state feedback with a deterministic gain achieves exact null control at every prescribed $T > 0$, with finite energy and a polynomial small-time bound.

For scalar noise and a surjective diffusion-control matrix, Liu and Peng [15, Definition 8 and Theorem 9] prove that, for the equivalent system with freely chosen diffusion, exact controllability is equivalent to arbitrary exponential convergence-index assignment by linear state feedback. Our classification also covers arbitrary joint diffusion rank, where ENC need not imply EC.

Sîrbu and Tessitore [20, Theorems 3.4 and 4.2] characterize exact null controllability in Hilbert spaces by a singular Riccati equation and a dual observability inequality, and obtain optimal null controls in feedback form. On the common finite-dimensional, constant-coefficient class, our result gives a finite algebraic test for this property. Their model uses independent Brownian families for the state and control noises; (1.1) also allows these terms to share the same Brownian channels.

For scalar noise and rank $D_1 = n$, the matrices E_1, F, H defined before Theorem 1.6 express Peng’s reduced criterion [19, Theorem 3.2], with the finite truncation of Lü and Zhang [16, Theorem 6.10], directly in A, B, C_1, D_1 without choosing a basis of $\ker D_1$. For the original system considered in their treatment, Lü and Zhang establish sufficiency but leave necessity open, and ask in [16, Remark 6.11] for a rank characterization directly in the original coefficients. Theorem 1.6 gives the requested formulation and also covers multidimensional noise.

Sun [22, Propositions 3.2–3.3 and Theorem 3.12] gives finite-word, Gramian, and Hautus characterizations for backward-structured systems with freely chosen diffusion, where EC and ENC coincide. The rank-difference criterion in Corollary 1.5 additionally covers nonsurjective $\mathcal{D}$, for which the diffusion cannot be prescribed freely. Section 4 evaluates this test by equivalent orthogonal elimination with polynomial storage and arithmetic cost, without forming the exponentially large block matrices.

Goreac, Li and Zhang [12] study reduced-rank mean-field systems by allowing weaker control integrability or prescribing Gaussian terminal laws; our classification retains $\mathcal{U}_T = L^2_{\mathbb{F}}(0, T; \mathbb{R}^r)$ and the four controllability requirements in Definition 1.1 for the constant-coefficient system (1.1).

For broader background on stochastic controllability, we refer to [9, 26] for early algebraic criteria, [3] for probabilistic controllability of additive-noise systems, [17] for partial observation, and [5] for invariant-subspace criteria with drift-only control. Gramian and duality characterizations are developed in [14, 23], while other terminal requirements include finite-chaos approximation [8], exact output control [24], and partial controllability with terminal constraints [21].

The remainder of the paper is organized as follows. Section 2 proves the classification and feedback theorems. Section 3 proves the rank criteria and discusses their deterministic reduction and nonrobustness. Section 4 presents the orthogonal elimination algorithm and a numerical comparison of input configurations in penalized stochastic linear-quadratic (LQ) control. Section 5 concludes the paper.

2 Proofs of Theorems 1.2 and 1.3

We first prove the equivalence of AC, ANC and $\mathcal{V}_* = \{0\}$ in Subsection 2.1. We then prove Theorem 1.3, which supplies the implication to ENC needed for part (i) of Theorem 1.2.

The feedback construction proceeds in three steps. In Subsection 2.2, we use approximate null controls to obtain stabilizing feedbacks with any prescribed mean-square decay rate. In Subsection 2.3, a semialgebraic argument bounds their gains and Lyapunov condition numbers polynomially in that rate. In Subsection 2.4, we apply these feedbacks on intervals accumulating at T : the exponential state decay dominates the polynomial growth and makes the control energy summable, establishing Theorem 1.3. We then complete the proof of Theorem 1.2 in Subsection 2.4.

2.1 Approximate controllability and the obstruction space

We prove the geometric characterization of AC and ANC using the conditional-invariance approach of Goreac [11, Theorem 3].

Lemma 2.1. *At every $T > 0$, system (1.1) is approximately controllable if and only if it is approximately null controllable, and these properties hold if and only if $\mathcal{V}_* = \{0\}$.*

Proof. Fix $T > 0$ and define $M_T : \mathcal{U}_T \rightarrow L^2_{\mathcal{F}_T}(\Omega; \mathbb{R}^n)$ by $M_T u = y(T; 0, u)$. This linear map is bounded by (1.2), and (1.5) gives

$$M_T^* \eta = B^\top z + \sum_{j=1}^m D_j^\top Z_j,$$

where (z, Z) solves (1.4). By linearity of the control system (1.1), AC is equivalent to density of the range of M_T in $L^2_{\mathcal{F}_T}(\Omega; \mathbb{R}^n)$, or equivalently to $\ker M_T^* = \{0\}$.

Suppose that $\mathcal{V}_* = \{0\}$ and $M_T^* \eta = 0$. We show by induction that $z(t) \in \mathcal{V}_k$ for all $t \in [0, T]$, $\mathbb{P}$ -a.s. For $k = 0$ this is trivial. Assume it holds for some k . Since $z(t) \in \mathcal{V}_k$, its projection onto $\mathcal{V}_k^\perp$ vanishes. Hence the projection of (1.4) onto $\mathcal{V}_k^\perp$ has zero drift and zero martingale part. Consequently, the quadratic variation of the martingale part satisfies

$$\sum_{j=1}^m \int_0^T |\Pi_{\mathcal{V}_k^\perp} Z_j(t)|^2 dt = 0, \quad \mathbb{P}\text{-a.s.}$$

Thus $Z_j(t) \in \mathcal{V}_k$ for every j , while the drift condition is

$$A^\top z(t) + \sum_{j=1}^m C_j^\top Z_j(t) \in \mathcal{V}_k,$$

almost everywhere with respect to the product measure $dt \otimes d\mathbb{P}$. Together with $B^\top z + \sum_{j=1}^m D_j^\top Z_j = 0$, (1.6)

and (1.7) give $z(t) \in \mathcal{V}_{k+1}$ almost everywhere. Since z is continuous and $\mathcal{V}_{k+1}$ is closed, this inclusion holds for all $t \in [0, T]$, $\mathbb{P}$ -a.s. At $k = n$, we obtain $z = 0$ and hence $\eta = z(T) = 0$. Thus AC holds, and ANC follows by taking the target to be zero.

Conversely, suppose that $\mathcal{V}_* \neq \{0\}$. Choose $0 \neq v \in \mathcal{V}_*$ and matrices $\Gamma_1, \dots, \Gamma_m$ as in Lemma 1.1, and let z solve

$$\begin{cases} dz(t) = -\left(A^\top + \sum_{j=1}^m C_j^\top \Gamma_j\right) z(t) dt + \sum_{j=1}^m \Gamma_j z(t) dW^j(t), \\ z(0) = v. \end{cases}$$

The solution z belongs to $L^2_{\mathbb{F}}(\Omega; C([0, T]; \mathbb{R}^n))$ and remains in $\mathcal{V}_*$ by (1.8). Thus

$$Z = (\Gamma_1 z, \dots, \Gamma_m z) \in L^2_{\mathbb{F}}(0, T; (\mathbb{R}^n)^m)$$

and $\eta = z(T) \in L^2_{\mathcal{F}_T}(\Omega; \mathbb{R}^n)$. The pair (z, Z) solves (1.4) and satisfies $B^\top z + \sum_{j=1}^m D_j^\top Z_j = 0$. Taking $y_0 = v$ in (1.5) gives

$$\mathbb{E}\langle y(T; v, u), \eta \rangle = |v|^2 \quad \text{for every } u \in \mathcal{U}_T.$$

If ANC held, there would be controls $u_\ell \in \mathcal{U}_T$ with $y(T; v, u_\ell) \rightarrow 0$ in $L^2_{\mathcal{F}_T}(\Omega; \mathbb{R}^n)$. The Cauchy–Schwarz inequality would then force the left-hand side to tend to zero, contradicting $v \neq 0$. Hence ANC fails, which proves the converse implication. $\square$

2.2 Arbitrary-rate stabilizability

We first obtain stabilizing feedbacks with any prescribed mean-square decay rate. We write $H > 0$ (respectively, $H \geq 0$) when a symmetric matrix H is positive definite (respectively, positive semidefinite), and use $<, \leq$ analogously.

For the $m \times m$ identity matrix I_m and any matrix M , the Kronecker product is

$$I_m \otimes M = \begin{pmatrix} M & 0 & \cdots & 0 \\ 0 & M & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & M \end{pmatrix},$$

with m copies of M on the diagonal.

For $\rho > 0$, a symmetric $P \in \mathbb{R}^{n \times n}$, and $U \in \mathbb{R}^{r \times n}$, put

$$\mathcal{M}_\rho(P, U) = \begin{pmatrix} AP + PA^\top + BU + U^\top B^\top + \rho P & (\mathcal{C}P + \mathcal{D}U)^\top \\ \mathcal{C}P + \mathcal{D}U & -I_m \otimes P \end{pmatrix}. \quad (2.1)$$

The matrix $\mathcal{M}_\rho(P, U)$ has size $(m+1)n$. A pair (P, U) is called feasible at rate ρ if $\mathcal{M}_\rho(P, U) < 0$; its lower-right block then gives $P > 0$. Approximate null controllability gives feasibility at every positive rate.

Lemma 2.2. *If $\mathcal{V}_* = \{0\}$, then for every $\rho > 0$ there exist a symmetric matrix $P \in \mathbb{R}^{n \times n}$ and a matrix $U \in \mathbb{R}^{r \times n}$ such that $\mathcal{M}_\rho(P, U) < 0$.*

Proof. Step 1: a linear family of approximate null controls. Fix $\rho > 0$ and set $\delta = e^{-\rho}/2$, so that $e^\rho \delta = 1/2$. By Lemma 2.1, for each coordinate vector e_i there is a control $u^{(i)} \in \mathcal{U}_1$ whose state $y^{(i)}(t) = y(t; e_i, u^{(i)})$ satisfies

$$\mathbb{E}|y^{(i)}(1)|^2 \leq \frac{\delta}{n}, \quad i = 1, \dots, n.$$

For $x = (x_1, \dots, x_n)^\top \in \mathbb{R}^n$, define $u^x = \sum_{i=1}^n x_i u^{(i)}$. By linearity, its state is $y^x = \sum_{i=1}^n x_i y^{(i)}$, and the Cauchy-Schwarz inequality gives

$$\mathbb{E}|y^x(1)|^2 \leq |x|^2 \sum_{i=1}^n \mathbb{E}|y^{(i)}(1)|^2 \leq \delta |x|^2. \quad (2.2)$$

Each $u^{(i)}$ has finite energy, and (1.2) gives the same property for $y^{(i)}$. Set

$$C_\rho := \sum_{i=1}^n \mathbb{E} \int_0^1 (|y^{(i)}(s)|^2 + |u^{(i)}(s)|^2) ds < \infty.$$

Applying the Cauchy-Schwarz inequality to the sums defining y^x and u^x and integrating in time gives

$$\mathbb{E} \int_0^1 (|y^x(s)|^2 + |u^x(s)|^2) ds \leq |x|^2 \sum_{i=1}^n \mathbb{E} \int_0^1 (|y^{(i)}(s)|^2 + |u^{(i)}(s)|^2) ds = C_\rho |x|^2. \quad (2.3)$$

The controls $u^{(i)}$ are now fixed, so C_ρ is independent of x .

Step 2: restarting the controls and obtaining finite weighted cost. The Wiener-space representation in [7, Propositions 9–10] allows us to write each $u^{(i)}(s)$ as a nonanticipative measurable functional $\Phi_i(s, W)$ of the path in $C([0, 1]; \mathbb{R}^m)$, up to $ds \otimes d\mathbb{P}$ -null sets. For an integer $j \geq 0$, put $W_j(s) = W(j+s) - W(j)$, $0 \leq s \leq 1$, and define $u_j^{(i)}(s) = \Phi_i(s, W_j)$. Let $y_j^{(i)}$ be the corresponding state starting from e_i and driven by W_j . On $[j, j+1)$, use

$$u(j+s) = \sum_{i=1}^n y_i(j) u_j^{(i)}(s), \quad 0 \leq s < 1,$$

where $y_i(j)$ is the i th coordinate of the state already constructed at time j . Linearity gives

$$y(j+s) = \sum_{i=1}^n y_i(j) y_j^{(i)}(s), \quad 0 \leq s \leq 1.$$

Conditionally on $\mathcal{F}_j$, the coefficients $y_i(j)$ are fixed, whereas W_j is a standard m -dimensional Brownian motion independent of $\mathcal{F}_j$. The family $(y_j^{(i)}, u_j^{(i)})_{i=1}^n$ consequently has the same joint law as $(y^{(i)}, u^{(i)})_{i=1}^n$ and is independent of $\mathcal{F}_j$. Thus (2.2) gives

$$\mathbb{E}(|y(j+1)|^2 \mid \mathcal{F}_j) \leq \delta |y(j)|^2. \quad (2.4)$$

Similarly, (2.3) gives

$$\mathbb{E}\left(\int_j^{j+1} (|y(t)|^2 + |u(t)|^2) dt \mid \mathcal{F}_j\right) \leq C_\rho |y(j)|^2. \quad (2.5)$$

This constructs a progressively measurable control with finite energy on every bounded interval. Taking expectations in (2.4) and iterating gives

$$\mathbb{E}|y(j)|^2 \leq \delta^j |y_0|^2.$$

For the weighted cost on $[j, j+1]$, (2.5) now implies

$$\mathbb{E} \int_j^{j+1} e^{\rho t} (|y(t)|^2 + |u(t)|^2) dt \leq C_\rho e^{\rho(j+1)} \mathbb{E}|y(j)|^2 \leq C_\rho e^\rho (e^\rho \delta)^j |y_0|^2.$$

Since $e^\rho \delta = 1/2$, summing over $j \geq 0$ gives

$$\mathbb{E} \int_0^\infty e^{\rho t} (|y(t)|^2 + |u(t)|^2) dt \leq C_\rho e^\rho \sum_{j=0}^\infty 2^{-j} |y_0|^2 = 2C_\rho e^\rho |y_0|^2. \quad (2.6)$$

Step 3: from finite weighted cost to the matrix inequality. Set $X(t) = e^{\rho t/2} y(t)$ and $v(t) = e^{\rho t/2} u(t)$. The transformed system has drift matrix $A_\rho = A + \rho I_n/2$, and (2.6) bounds its unweighted infinite-horizon cost by $2C_\rho e^\rho |y_0|^2$. We use finite-horizon value matrices to obtain a stabilizing feedback, following the weighted LQ approach of [15].

For a symmetric matrix H , define

$$\mathbf{R}(H) = I_r + \sum_{j=1}^m D_j^\top H D_j$$

and the state-control coefficient

$$\mathbf{L}(H) = B^\top H + \sum_{j=1}^m D_j^\top H C_j.$$

Put

$$F_\rho(H) = A_\rho^\top H + H A_\rho + \sum_{j=1}^m C_j^\top H C_j + I_n - \mathbf{L}(H)^\top \mathbf{R}(H)^{-1} \mathbf{L}(H).$$

The map F_ρ is defined wherever $\mathbf{R}(H) > 0$. Consider the Riccati equation in the horizon length s ,

$$\begin{cases} \dot{\mathbf{H}}_\rho(s) = F_\rho(\mathbf{H}_\rho(s)), \\ \mathbf{H}_\rho(0) = 0. \end{cases} \quad (2.7)$$

With drift A_ρ , state and control weights $Q = I_n$ and $R = I_r$, zero cross weight, and zero terminal weight, the standard stochastic LQ theorem [25, Chapter 6, Theorem 7.2, p. 320] applies. Its multidimensional version is given by equation (7.60) and the discussion on pp. 334–335 therein. The constant coefficients and weights satisfy its boundedness, continuity and positivity assumptions. After time reversal, it gives a unique nonnegative solution of (2.7) on every bounded interval. By uniqueness, these solutions agree on overlapping intervals and define a global solution satisfying $R(H_\rho(s)) \geq I_r$ for $s \geq 0$. Moreover, $\dot{H}_\rho(0) = I_n$. Fix $s > 0$ and set $H_t = H_\rho(s - t)$. For $X(0) = x$, Itô's formula and completion of squares give

$$\mathbb{E} \int_0^s (|X(t)|^2 + |v(t)|^2) dt = x^\top H_\rho(s)x + \mathbb{E} \int_0^s |R(H_t)^{1/2}v + R(H_t)^{-1/2}L(H_t)X|^2 dt.$$

The last integral is nonnegative and vanishes for the feedback displayed below. Consequently,

$$x^\top H_\rho(s)x = \inf_v \mathbb{E} \int_0^s (|X(t)|^2 + |v(t)|^2) dt,$$

where $X(0) = x$ and the infimum is over square-integrable progressively measurable controls on $[0, s]$. It is attained by

$$v(t) = -R(H_\rho(s - t))^{-1}L(H_\rho(s - t))X(t).$$

Nonnegative running cost makes $H_\rho(s)$ nondecreasing in the Loewner order. Restricting the control from (2.6) to $[0, s]$ gives

$$0 \leq H_\rho(s) \leq 2C_\rho e^\rho I_n.$$

Consequently, $H_\rho(s)$ converges to a symmetric matrix H_ρ . For fixed $h > 0$, the integral form of (2.7) gives

$$H_\rho(s + h) - H_\rho(s) = \int_0^h F_\rho(H_\rho(s + \tau)) d\tau.$$

Letting $s \rightarrow \infty$ and using continuity of F_ρ gives $F_\rho(H_\rho) = 0$. Since $\dot{H}_\rho(0) = I_n$ and the value matrices are nondecreasing, $H_\rho > 0$.

Set $K_\rho = -R(H_\rho)^{-1}L(H_\rho)$. The identity $F_\rho(H_\rho) = 0$ becomes

$$(A + BK_\rho)^\top H_\rho + H_\rho(A + BK_\rho) + \rho H_\rho + \sum_{j=1}^m (C_j + D_j K_\rho)^\top H_\rho (C_j + D_j K_\rho) = -I_n - K_\rho^\top K_\rho < 0. \quad (2.8)$$

Take $P = H_\rho^{-1}$ and $U = K_\rho P$, so that $(C_j + D_j K_\rho)P = C_j P + D_j U$. Multiplying (2.8) on both sides by P gives

$$AP + PA^\top + BU + U^\top B^\top + \rho P + (CP + DU)^\top (I_m \otimes P^{-1})(CP + DU) < 0.$$

Since $P > 0$, the Schur complement with lower-right block $-I_m \otimes P$ yields $\mathcal{M}_\rho(P, U) < 0$. $\square$

2.3 Polynomially bounded stabilizing feedbacks

We now seek feasible pairs for which the feedback gain and the condition number $\kappa(P) = \lambda_{\max}(P)/\lambda_{\min}(P)$ grow at most polynomially with the decay rate.

For this purpose, we use semialgebraic sets, defined by finite Boolean combinations of polynomial equalities and inequalities with real coefficients. A function is semialgebraic if its graph is semialgebraic. The Tarski–Seidenberg theorem permits projection and elimination of real quantifiers [4, Theorem 2.2.1 and Proposition 2.2.4].

Lemma 2.3. *The image of a semialgebraic subset of $\mathbb{R}^{a+b}$ under coordinate projection onto $\mathbb{R}^a$ is semialgebraic. More generally, a set defined by polynomial equalities and inequalities, finite Boolean operations and finitely many real existential or universal quantifiers is semialgebraic.*

We also use the polynomial growth bound for continuous semialgebraic functions on closed semialgebraic sets [4, Proposition 2.6.2, p. 43], in the following form.

Lemma 2.4. *If $f : [1, \infty) \rightarrow \mathbb{R}$ is continuous and semialgebraic, then there exist $M \geq 1$ and an integer $p \geq 1$ such that*

$$|f(\rho)| \leq M\rho^p, \quad \rho \geq 1.$$

Related real-algebraic methods have been used in output-feedback and stabilization problems [1, 10]. We apply these lemmas to the minimum of $\|P\|_F^2 + \|U\|_F^2$ over a normalized feasible set for (2.1). The normalization bounds P^{-1} , while the minimum bounds both $\kappa(P)$ and $\|UP^{-1}\|^2$.

Lemma 2.5. *Suppose that a feasible pair exists for every $\rho \geq 1$. There exist a constant $M \geq 1$ and an integer $p \geq 1$, depending only on $A, B, (C_j, D_j)_{j=1}^m$, such that for every $\rho \geq 1$ one can choose a feasible pair $(P(\rho), U(\rho))$ satisfying*

$$\kappa(P(\rho)) \leq M\rho^p \tag{2.9}$$

and

$$\|U(\rho)P(\rho)^{-1}\|^2 \leq M\rho^p. \tag{2.10}$$

Proof. Step 1: a semialgebraic minimum. For $\rho \geq 1$, define the closed set

$$\mathcal{F}_\rho = \{(P, U) : P = P^\top, \mathcal{M}_\rho(P, U) \leq -I_{(m+1)n}\}$$

and put $g(P, U) = \|P\|_F^2 + \|U\|_F^2$. By the feasibility assumption and the homogeneity

$$\mathcal{M}_\rho(cP, cU) = c\mathcal{M}_\rho(P, U), \quad c > 0,$$

$\mathcal{F}_\rho$ is nonempty. Its lower-right block gives $I_m \otimes P \geq I_{mn}$, hence $P \geq I_n$. Since g is coercive, it attains the finite minimum

$$\mu(\rho) = \min_{(P, U) \in \mathcal{F}_\rho} g(P, U).$$

We first show that μ is semialgebraic. The set of triples (ρ, P, U) with $\rho \geq 1$ and $(P, U) \in \mathcal{F}_\rho$ is semialgebraic: the matrix inequality is equivalent to nonnegativity of all principal minors of $-\mathcal{M}_\rho(P, U) - I_{(m+1)n}$. For $\rho \geq 1$, a point (ρ, s) belongs to the graph of μ precisely when some $(P, U) \in \mathcal{F}_\rho$ satisfies $g(P, U) = s$ and no $(P', U') \in \mathcal{F}_\rho$ satisfies $g(P', U') < s$. By Lemma 2.3, this graph is semialgebraic.

Step 2: continuity of the minimum. Let $\rho_j \rightarrow \rho$ in $[1, \infty)$. To prove lower semicontinuity, it suffices to consider $\liminf_j \mu(\rho_j) < \infty$. Pass to a subsequence realizing this lower limit and choose minimizing pairs $(P_j, U_j) \in \mathcal{F}_{\rho_j}$. Coercivity gives a further subsequence converging to a pair (P, U) . Passing to the limit in the matrix inequality gives $(P, U) \in \mathcal{F}_\rho$, and hence

$$\mu(\rho) \leq g(P, U) = \liminf_{j \rightarrow \infty} \mu(\rho_j).$$

For upper semicontinuity, choose a minimizing pair (P, U) at ρ and fix $c > 1$. For ρ' near ρ ,

$$\mathcal{M}_{\rho'}(cP, cU) = c\mathcal{M}_\rho(P, U) + c(\rho' - \rho) \begin{pmatrix} P & 0 \\ 0 & 0_{mn} \end{pmatrix} \leq -cI_{(m+1)n} + c(\rho' - \rho) \begin{pmatrix} P & 0 \\ 0 & 0_{mn} \end{pmatrix}.$$

Thus $(cP, cU) \in \mathcal{F}_{\rho'}$ whenever $\rho' \geq 1$ is sufficiently close to ρ , so

$$\limsup_{\rho' \rightarrow \rho} \mu(\rho') \leq c^2 \mu(\rho).$$

Letting $c \downarrow 1$ proves upper semicontinuity.

Step 3: polynomial bounds. Applying Lemma 2.4 to the continuous semialgebraic function $\mu : [1, \infty) \rightarrow \mathbb{R}$, we obtain a constant $M \geq 1$ and an integer $p \geq 1$ such that

$$\mu(\rho) \leq M\rho^p, \quad \rho \geq 1.$$

For any minimizing pair $(\bar{P}, \bar{U})$, we have $\bar{P} \geq I_n$ and $\mu(\rho) \geq n \geq 1$. In particular, $\|\bar{P}^{-1}\| \leq 1$, so

$$\kappa(\bar{P}) \leq \|\bar{P}\|_F \leq \sqrt{\mu(\rho)} \leq \mu(\rho).$$

The gain satisfies

$$\|\bar{U}\bar{P}^{-1}\|^2 \leq \|\bar{U}\|_F^2 \leq \mu(\rho).$$

Taking $P(\rho) = \bar{P}$ and $U(\rho) = \bar{U}$ proves (2.9) and (2.10). $\square$

We next record the decay estimate that will be used on each switching interval. For a feasible pair (P, U) at rate ρ , set $K = UP^{-1}$. The Schur complement of (2.1), followed by multiplication on both sides by P^{-1} , gives

$$(A + BK)^\top P^{-1} + P^{-1}(A + BK) + \rho P^{-1} + \sum_{j=1}^m (C_j + D_j K)^\top P^{-1}(C_j + D_j K) < 0. \quad (2.11)$$

Define $V(x) = \langle P^{-1}x, x \rangle$ and apply the feedback $u = Ky$. The initial state $y(t)$ may be any square-integrable $\mathcal{F}_t$ -measurable vector. Itô's formula gives

$$\begin{aligned} dV(y(\tau)) &= 2\langle P^{-1}y(\tau), (A + BK)y(\tau) \rangle d\tau + \sum_{j=1}^m |P^{-1/2}(C_j + D_j K)y(\tau)|^2 d\tau \\ &\quad + 2 \sum_{j=1}^m \langle P^{-1}y(\tau), (C_j + D_j K)y(\tau) \rangle dW^j(\tau). \end{aligned}$$

By (2.11), the drift is at most $-\rho V(y(\tau))$. Multiplying by $e^{\rho(\tau-t)}$, localizing the stochastic integral, and taking expectations yields

$$\mathbb{E}V(y(t+s)) \leq e^{-\rho s} \mathbb{E}V(y(t)).$$

Finally, the eigenvalue bounds

$$\frac{|x|^2}{\lambda_{\max}(P)} \leq V(x) \leq \frac{|x|^2}{\lambda_{\min}(P)}$$

give

$$\mathbb{E}|y(t+s)|^2 \leq \kappa(P)e^{-\rho s} \mathbb{E}|y(t)|^2. \quad (2.12)$$

Both K and P are deterministic, so (2.12) applies without changing the feedback when $y(t)$ is random.

2.4 Feedback construction and exact controllability

Proof of Theorem 1.3. Assume $\mathcal{V}_* = \{0\}$. We first consider $0 < T \leq 1$ and treat $T > 1$ at the end of the construction. By Lemmas 2.2 and 2.5, we may choose feedbacks satisfying (2.9) and (2.10) at every rate $\rho \geq 1$. We will shorten the switching intervals while increasing the decay rates fast enough that the factor $e^{-\rho s}$ in (2.12) compensates for both polynomial factors in (2.9) and (2.10).

Step 1: choosing the intervals and decay rates. Choose $L \geq \max(1, 4p)$ so large that

$$4^{2p} M L^p e^{-L/2} \leq \frac{1}{2},$$

which is possible because $e^{-L/2}$ decays faster than L^p . Set

$$t_j = T(1 - 2^{-j}), \quad j \geq 0. \quad (2.13)$$

The interval lengths are $\ell_j = t_{j+1} - t_j = T2^{-j-1}$. Choose $\lambda_j = LT^{-2}4^j$ and $K_j = U(\lambda_j)P(\lambda_j)^{-1}$, and define

$$K_T(t) = K_j, \quad t \in [t_j, t_{j+1}), \quad j \geq 0.$$

In particular, $K_T(t_j) = K_j$ at every switching time. On each interval, we solve the linear SDE with drift matrix $A + BK_j$ and diffusion matrices $C_i + D_i K_j$, $i = 1, \dots, m$, using the terminal state from the preceding interval as its initial state, starting with $y(0) = y_0$. By [16, Theorem 3.2, p. 132], these equations have unique square-integrable continuous adapted solutions. They therefore join continuously at the switching times. Since K_T is bounded on every $[0, s]$ with $s < T$, this gives a unique continuous adapted state on $[0, T)$, with $u(t) = K_T(t)y(t)$ locally square-integrable. Although $\ell_j \rightarrow 0$, the products $\lambda_j \ell_j = L2^{j-1}/T$ tend to infinity. Writing $a_j = \mathbb{E}|y(t_j)|^2$, (2.9) and (2.12) give

$$a_{j+1} \leq M\lambda_j^p e^{-\lambda_j \ell_j} a_j. \quad (2.14)$$

Step 2: summability of the control energy. The energy on $[t_j, t_{j+1}]$ contains the product of $\|K_j\|^2$ and $\kappa(P(\lambda_j))$. Indeed, (2.9), (2.10) and (2.12) imply

$$\mathbb{E} \int_{t_j}^{t_{j+1}} |u(t)|^2 dt \leq \|K_j\|^2 \kappa(P(\lambda_j)) a_j \int_0^{\ell_j} e^{-\lambda_j s} ds \leq M^2 \lambda_j^{2p} a_j \int_0^{\ell_j} e^{-\lambda_j s} ds \leq M^2 \lambda_j^{2p-1} a_j. \quad (2.15)$$

This motivates the definition $b_j = \lambda_j^{2p} a_j$, for which (2.15) is bounded by $M^2 b_j / \lambda_j$. Since $\lambda_{j+1} = 4\lambda_j$, (2.14) gives

$$b_{j+1} = \lambda_{j+1}^{2p} a_{j+1} \leq 4^{2p} M \lambda_j^p e^{-\lambda_j \ell_j} b_j \leq \frac{1}{2} b_j. \quad (2.16)$$

To verify the last inequality, put $x_j = 2^j / T \geq 1$. Then

$$4^{2p} M \lambda_j^p e^{-\lambda_j \ell_j} = 4^{2p} M L^p x_j^{2p} e^{-Lx_j/2} \leq 4^{2p} M L^p e^{-L/2} \leq \frac{1}{2}.$$

Here $x \mapsto x^{2p} e^{-Lx/2}$ is nonincreasing on $[1, \infty)$ because $L \geq 4p$. It follows from (2.16) that $b_j \leq 2^{-j} b_0$. This, together with $\lambda_j = 4^j \lambda_0$, gives

$$\frac{b_j}{\lambda_j} \leq 8^{-j} \frac{b_0}{\lambda_0}.$$

Summing (2.15), we therefore obtain

$$\mathbb{E} \int_0^T |u(t)|^2 dt \leq M^2 \sum_{j=0}^{\infty} \frac{b_j}{\lambda_j} \leq \frac{M^2 b_0}{\lambda_0} \sum_{j=0}^{\infty} 8^{-j} = \frac{8M^2 b_0}{7\lambda_0} = \frac{8}{7} M^2 L^{2p-1} T^{-(4p-2)} |y_0|^2. \quad (2.17)$$

In the last equality we used $b_0 = \lambda_0^{2p} |y_0|^2$ and $\lambda_0 = LT^{-2}$.

Step 3: extending the state to the terminal time. So far, (y, u) has been constructed on $[0, T)$ by solving the feedback equation on successive intervals. Set $u(T) = 0$; this single endpoint value does not affect the integrals. The resulting control is progressively measurable, and (2.17) gives $u \in \mathcal{U}_T$. Regarding u as a fixed admissible control, we apply [16, Theorem 3.2, p. 132] to (1.1) on $[0, T]$. The resulting solution belongs to $L^2_{\mathbb{F}}(\Omega; C([0, T]; \mathbb{R}^n))$ and, by uniqueness, agrees with the successively constructed state on every $[0, t_j]$. We retain the notation y for this extension. Any other continuous adapted closed-loop solution agrees with y on every $[0, t_j]$ by local uniqueness, and hence also at T by continuity. Since $\lambda_j \geq 1$, (2.16) implies

$$a_j = \lambda_j^{-2p} b_j \leq 2^{-j} b_0 \rightarrow 0.$$

In particular, y is continuous in $L^2(\Omega; \mathbb{R}^n)$ at T . Consequently,

$$\mathbb{E}|y(T)|^2 = \lim_{j \rightarrow \infty} \mathbb{E}|y(t_j)|^2 = \lim_{j \rightarrow \infty} a_j = 0.$$

The deterministic gain schedule is independent of y_0 , so the resulting null-control map is linear and bounded. Together with (2.17), the identity $y(T) = 0$ proves the theorem for $0 < T \leq 1$, with $N = 4p - 2$ in (1.10).

For $T > 1$, let $\hat{K}$ be the feedback schedule constructed above for the unit horizon. Set $K_T(t) = 0$ on $[0, T-1)$ and $K_T(t) = \hat{K}(t - T + 1)$ on $[T-1, T)$. The uncontrolled state $y(T-1)$ is square-integrable. Since (2.12) applies to square-integrable $\mathcal{F}_{T-1}$ -measurable initial states, the preceding estimates apply on the last unit interval with $|y_0|^2$ replaced by $\mathbb{E}|y(T-1)|^2$. They give finite control energy and $y(T) = 0$. This schedule is locally bounded on $[0, T)$ and independent of y_0 , completing the proof. $\square$

Proof of Theorem 1.2, part (i). By Lemma 2.1, AC and ANC are each equivalent to $\mathcal{V}_* = \{0\}$. By Theorem 1.3, this condition implies ENC. Conversely, ENC implies ANC, completing the equivalence. $\square$

Proof of Theorem 1.2, part (ii). Fix $T > 0$. Exact controllability implies approximate null controllability and hence $\mathcal{V}_* = \{0\}$. It also makes the map $M_T : \mathcal{U}_T \rightarrow L^2_{\mathcal{F}_T}(\Omega; \mathbb{R}^n)$, $M_T u = y(T; 0, u)$, surjective. Hilbert-space duality therefore gives a constant $c(T) > 0$ such that every adjoint solution satisfies

$$\mathbb{E}|\eta|^2 \leq c(T) \mathbb{E} \int_0^T \left| B^\top z(t) + \sum_{j=1}^m D_j^\top Z_j(t) \right|^2 dt. \quad (2.18)$$

We show that this requires $\text{rank } \mathcal{D} = mn$. Otherwise, choose $q = [q_1^\top \cdots q_m^\top]^\top \in \ker \mathcal{D}^\top$ with $|q|^2 = \sum_{j=1}^m |q_j|^2 = 1$. For $0 < \varepsilon < \min(T, 1)$, prescribe

$$Z_j^\varepsilon(t) = \varepsilon^{-1/2} q_j \mathbf{1}_{[T-\varepsilon, T]}(t), \quad j = 1, \dots, m,$$

solve the adjoint differential equation forward from $z^\varepsilon(0) = 0$, and put $\eta_\varepsilon = z^\varepsilon(T)$. By uniqueness, this is the adjoint solution for terminal value η_ε . Set $g = \sum_{j=1}^m C_j^\top q_j$. Then $z^\varepsilon(t) = 0$ for $t \leq T - \varepsilon$, and for later times

$$z^\varepsilon(t) = -\varepsilon^{-1/2} \int_{T-\varepsilon}^t e^{-A^\top(t-s)} g ds + \varepsilon^{-1/2} \sum_{j=1}^m \int_{T-\varepsilon}^t e^{-A^\top(t-s)} q_j dW^j(s).$$

The Itô isometry gives

$$\mathbb{E}|\eta_\varepsilon|^2 = \frac{1}{\varepsilon} \sum_{j=1}^m \int_0^\varepsilon |e^{-A^\top s} q_j|^2 ds + \frac{1}{\varepsilon} \left| \int_0^\varepsilon e^{-A^\top s} g ds \right|^2 \longrightarrow 1.$$

On the other hand, $\sum_{j=1}^m D_j^\top Z_j^\varepsilon = 0$. Let $M_A = \max_{0 \leq s \leq 1} \|e^{-A^\top s}\|$ and $c_* = M_A^2(1 + \sum_{j=1}^m \|C_j\|^2)$. For $\tau = t - (T - \varepsilon) \in [0, \varepsilon]$, the deterministic term in the representation of $z^\varepsilon(t)$ has zero cross expectation with the stochastic integral. The Itô isometry and the Cauchy-Schwarz inequality therefore give

$$\mathbb{E}|z^\varepsilon(t)|^2 = \frac{1}{\varepsilon} \left| \int_0^\tau e^{-A^\top s} g ds \right|^2 + \frac{1}{\varepsilon} \sum_{j=1}^m \int_0^\tau |e^{-A^\top s} q_j|^2 ds \leq \frac{M_A^2}{\varepsilon} \left(\tau^2 |g|^2 + \tau \sum_{j=1}^m |q_j|^2 \right) \leq \frac{c_*}{\varepsilon} (\tau^2 + \tau) \leq 2c_*.$$

Here we used $|g|^2 \leq \sum_{j=1}^m \|C_j\|^2$, $\sum_{j=1}^m |q_j|^2 = 1$, and $0 \leq \tau \leq \varepsilon < 1$. In particular, c_* is independent of ε . Consequently,

$$\lim_{\varepsilon \rightarrow 0} \mathbb{E} \int_0^T \left| B^\top z^\varepsilon(t) + \sum_{j=1}^m D_j^\top Z_j^\varepsilon(t) \right|^2 dt = \lim_{\varepsilon \rightarrow 0} \mathbb{E} \int_{T-\varepsilon}^T |B^\top z^\varepsilon(t)|^2 dt \leq \lim_{\varepsilon \rightarrow 0} 2c_* \|B\|^2 \varepsilon = 0.$$

Together with $\mathbb{E}|\eta_\varepsilon|^2 \rightarrow 1$, this contradicts (2.18).

Conversely, suppose that $\mathcal{V}_* = \{0\}$ and $\mathcal{D}$ is onto. Set $J = \mathcal{D}^\top(\mathcal{D}\mathcal{D}^\top)^{-1}$. For $\xi \in L^2_{\mathcal{F}_T}(\Omega; \mathbb{R}^n)$, let $(x, \zeta_1, \dots, \zeta_m)$ solve

$$\begin{cases} dx(t) = ((A - BJC)x(t) + BJ\zeta(t)) dt + \sum_{j=1}^m \zeta_j(t) dW^j(t), \\ x(T) = \xi. \end{cases}$$

where $\zeta = [\zeta_1^\top \dots \zeta_m^\top]^\top$. By [16, Theorem 4.2], this BSDE has a square-integrable solution. Define $\hat{u} = J(\zeta - \mathcal{C}x) \in \mathcal{U}_T$. Since $\mathcal{D}J = I_{mn}$, its diffusion coefficients satisfy

$$\mathcal{C}x + \mathcal{D}\hat{u} = \zeta.$$

The drift coefficient is

$$Ax + B\hat{u} = (A - BJC)x + BJ\zeta.$$

Thus $(x, \hat{u})$ solves (1.1) with terminal state ξ . Since $\mathcal{F}_0$ is trivial modulo null sets, $x(0)$ is deterministic. Theorem 1.3 supplies a control u^0 whose trajectory y^0 starts from $y_0 - x(0)$ and satisfies $y^0(T) = 0$. By linearity, $y = x + y^0$ is the state corresponding to $u = \hat{u} + u^0$, with initial state

$$y(0) = x(0) + (y_0 - x(0)) = y_0.$$

Its terminal state is $y(T) = x(T) + y^0(T) = \xi$. This proves (1.9), and hence part (ii). $\square$

Remark 2.1. In the terminology of [15], exact terminal controllability allows a deterministic initial state x_ξ to be chosen together with the control for each target $\xi \in L^2_{\mathcal{F}_T}(\Omega; \mathbb{R}^n)$, rather than prescribing the initial state. The BSDE construction above produces such an $x_\xi = x(0)$ when $\mathcal{D}$ is onto. Exact null controllability then corrects the difference $y_0 - x_\xi$ without changing the terminal target. Thus, by linearity, exact controllability is equivalent to exact terminal controllability together with exact null controllability. Here x_ξ is deterministic, not a random initial state depending on future noise.

3 Proofs of Theorems 1.4 and 1.6

We now prove the criteria in the original coefficients stated in Theorems 1.4 and 1.6.

Proof of Theorem 1.4. We prove (1.14) by induction. For $k = 1$, the drift condition in (1.6) is unrestricted, so (1.11) gives the representation. At step $k + 1$, the conditions $w_j \in \mathcal{V}_k$ are represented by

$$\mathbf{K}_k w_j + \mathbf{N}_k \alpha_j = 0, \quad j = 1, \dots, m. \quad (3.1)$$

The drift constraint is

$$\mathbf{K}_k \left(A^\top v + \sum_{j=1}^m C_j^\top w_j \right) + \mathbf{N}_k \beta = 0. \quad (3.2)$$

Combining (3.1) and (3.2) with $B^\top v + \sum_{j=1}^m D_j^\top w_j = 0$, we obtain

$$\mathbf{K}_{k+1} v + \mathbf{N}_{k+1} [w_1^\top \dots w_m^\top \alpha_1^\top \dots \alpha_m^\top \beta^\top]^\top = 0,$$

with the block matrices from (1.12).

Conversely, the block equation imposes the zero-observation constraint and places every w_j and the drift $A^\top v + \sum_{j=1}^m C_j^\top w_j$ in $\mathcal{V}_k$. These conditions also imply $v \in \mathcal{V}_k$. Indeed, starting with $v \in \mathcal{V}_0$, if $v \in \mathcal{V}_\ell$ for $0 \leq \ell < k$, then $\mathcal{V}_k \subset \mathcal{V}_\ell$ and (1.6) and (1.7) give $v \in \mathcal{V}_{\ell+1}$. The same constraints therefore yield $v \in \mathcal{V}_{k+1}$, proving (1.14).

For matrices K, N with the same number of rows, the linear map $v \mapsto Kv + \text{im } N$ into the quotient by $\text{im } N$ has kernel $\{v : Kv \in \text{im } N\}$ and rank $(\text{rank}[N \ K] - \text{rank } N)$. This proves (1.15). $\square$

Every block is polynomial in $A, B, (C_j, D_j)_{j=1}^m$, so the criterion does not require computing the obstruction spaces or inverting a coefficient matrix.

Remark 3.1. When the diffusion coefficients vanish, Corollary 1.5 recovers the classical Kalman rank condition. The distinction between deterministic and random terminal targets explains why EC does not become equivalent to the other three properties in this case.

Indeed, if $C_j = D_j = 0$ for every j , the recursion in (1.6) gives

$$\mathcal{V}_k = \bigcap_{\ell=0}^{k-1} \ker\{B^\top (A^\top)^\ell\} \quad (k \geq 1).$$

Hence

$$\delta_n = \text{rank}[B \ AB \ \cdots \ A^{n-1}B]. \quad (3.3)$$

Consequently, (1.16) becomes the Kalman rank condition for AC, ANC and ENC. For the classical problem with deterministic controls and terminal targets, this condition is equivalent to all four controllability properties. Here controls remain in $\mathcal{U}_T$ and terminal targets range over $L^2_{\mathcal{F}_T}(\Omega; \mathbb{R}^n)$: adapted controls can produce random terminal states even when the diffusion vanishes. Exact controllability to all such targets fails because $\mathcal{D} = 0$ violates the necessary condition in Corollary 1.5.

The following construction explains how adapted controls approximate arbitrary random terminal targets even when the diffusion vanishes. Assume the Kalman condition. For $\xi \in L^2_{\mathcal{F}_T}(\Omega; \mathbb{R}^n)$ and $0 < h < T$, set $\xi_h = \mathbb{E}[\xi \mid \mathcal{F}_{T-h}]$. The target ξ_h is already known at time $T - h$. After zero control on $[0, T - h]$, the deterministic Gramian formula gives a control on $[T - h, T]$ reaching ξ_h exactly. This control is adapted and has finite expected energy for each h , since ξ_h is $\mathcal{F}_{T-h}$ -measurable and square-integrable. Under the Brownian natural filtration, $\xi_h \rightarrow \xi$ in $L^2_{\mathcal{F}_T}(\Omega; \mathbb{R}^n)$ as $h \downarrow 0$, which gives AC. The corresponding control energies need not remain bounded.

Remark 3.2. The first iterate is

$$\mathcal{V}_1 = (B \ker \mathcal{D})^\perp. \quad (3.4)$$

If $B \ker \mathcal{D} = \{0\}$, the iteration is stationary from the start. Hence each of the four controllability properties requires $B \ker \mathcal{D} \neq \{0\}$. Exact controllability also requires $\text{rank } \mathcal{D} = mn$, so the control dimension satisfies $r \geq mn + 1$. Thus, besides enough input directions to prescribe all diffusion channels, at least one input combination that vanishes in every diffusion channel must act nontrivially through the drift.

The following example, adapted from [16, Section 6.3 and Remark 6.24], shows that arbitrarily small diffusion perturbations can destroy AC, ANC and ENC, ruling out a full-rank characterization of these properties by a fixed-size matrix depending continuously on the coefficients.

Example 3.1. For fixed $m \geq 1$, take $n = 2$, $r = 1$ and consider

$$\begin{cases} dy_1 = y_2 dt + \varepsilon y_2 dW^1(t), \\ dy_2 = u dt. \end{cases}$$

Its drift coefficients are $B = (0, 1)^\top$ and

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

The diffusion coefficients are $C_1 = \varepsilon A$, $C_j = 0$ for $j \geq 2$, and $D_j = 0$ for every j . Since $B^\top v = v_2$, the first step of (1.7) gives $\mathcal{V}_1 = \text{span}\{e_1\}$ for every ε . At the second step, write $v = ae_1$ and $w_j = b_j e_1$. The remaining constraint is

$$A^\top v + \sum_{j=1}^m C_j^\top w_j = (a + \varepsilon b_1)e_2 \in \text{span}\{e_1\},$$

or equivalently $a + \varepsilon b_1 = 0$. If $\varepsilon = 0$, this forces $a = 0$; if $\varepsilon \neq 0$, every a is admissible by choosing $b_1 = -a/\varepsilon$ and $b_j = 0$ for $j \geq 2$. Since $n = 2$, it follows that

$$\mathcal{V}_* = \mathcal{V}_2 = \begin{cases} \{0\}, & \varepsilon = 0, \\ \text{span}\{e_1\}, & \varepsilon \neq 0. \end{cases}$$

By Theorem 1.2, AC, ANC and ENC hold at every $T > 0$ when $\varepsilon = 0$, and all three fail at every $T > 0$ when $\varepsilon \neq 0$. Exact controllability fails throughout this family because $\mathcal{D} = 0$.

For fixed n, r, m , maximal rank of a fixed-size matrix depending continuously on the coefficient matrices is an open condition. The family above therefore precludes such a full-rank characterization of AC, ANC or ENC in general. For the rank-difference criterion, (1.15) gives $\delta_2 = 2$ when $\varepsilon = 0$ and $\delta_2 = 1$ otherwise. Thus the condition $\delta_2 = n$ in (1.16) is not open along this family.

Proof of Theorem 1.6. Put $w = [w_1^\top \cdots w_m^\top]^\top$. Since $J\mathcal{D}$ is the orthogonal projector onto $\text{im } \mathcal{D}^\top$, we have $(I_r - J\mathcal{D})\mathcal{D}^\top = 0$ and $J^\top \mathcal{D}^\top = I_{mn}$. Multiplying $B^\top v + \mathcal{D}^\top w = 0$ by $I_r - J\mathcal{D}$ and $J^\top$ therefore gives

$$H^\top v = 0, \quad w = -J^\top B^\top v.$$

Conversely, these conditions imply

$$B^\top v + \mathcal{D}^\top w = (I_r - \mathcal{D}^\top J^\top)B^\top v = H^\top v = 0.$$

The second condition reads $w_j = -E_j^\top v$ for every j , and the drift becomes

$$A^\top v + \sum_{j=1}^m C_j^\top w_j = (A^\top - C^\top J^\top B^\top)v = F^\top v.$$

By Lemma 1.1, $\mathcal{V}_*$ is therefore the largest common $(E_1^\top, \dots, E_m^\top, F^\top)$ -invariant subspace contained in $\ker H^\top$.

Put $S_k = \text{im}[H_0 \ H_1 \ \cdots \ H_k]$. By (1.17),

$$S_{k+1} = S_k + \sum_{j=1}^m E_j S_k + F S_k.$$

Thus the sequence is increasing, and an equality $S_{k+1} = S_k$ implies invariance under every E_j and F , so all subsequent spaces coincide. If $H = 0$, all these spaces are zero. Otherwise $\dim S_0 \geq 1$; if no equality occurs before S_{n-1} , then $\dim S_{n-1} \geq \dim S_0 + n - 1 \geq n$. In either case, $S := S_{n-1}$ is invariant under all these matrices. If U is any other common invariant subspace containing $\text{im } H$, the recursion gives $S_k \subset U$ by induction. Hence S is the smallest such subspace.

For $v \in S^\perp$, $x \in S$, and $M \in \{E_1, \dots, E_m, F\}$, we have

$$\langle M^\top v, x \rangle = \langle v, Mx \rangle = 0.$$

Consequently, $S^\perp$ is invariant under $E_1^\top, \dots, E_m^\top, F^\top$. Since $\text{im } H \subset S$, it is also contained in $\ker H^\top$. The maximality of $\mathcal{V}_*$ therefore gives $S^\perp \subset \mathcal{V}_*$. Conversely, the same inner-product identity shows that $\mathcal{V}_*^\perp$ is invariant under every E_j and F , and $\mathcal{V}_* \subset \ker H^\top$ implies $\text{im } H \subset \mathcal{V}_*^\perp$. The minimality of S gives $S \subset \mathcal{V}_*^\perp$, and hence $\mathcal{V}_* \subset S^\perp$. We conclude that $S_{n-1}^\perp = \mathcal{V}_*$. Thus (1.18) is equivalent to $\mathcal{V}_* = \{0\}$, and the conclusion follows from Theorem 1.2. $\square$

Remark 3.3. Since $I_r - J\mathcal{D}$ is the orthogonal projector onto $\ker \mathcal{D}$,

$$\text{im } H = B \ker \mathcal{D}.$$

Thus H represents the drift-control directions left after the diffusion has been prescribed, without choosing a basis of $\ker \mathcal{D}$. For $m = 1$, the matrices E_1, F, H express Peng's reduced word criterion [19, Theorem 3.2] directly in the original coefficients, with the finite truncation in [16, Theorem 6.10]. Related finite-word characterizations for backward-structured systems are given in [22, Propositions 3.2–3.3].

The block dimensions in (1.12) grow exponentially. Section 4 gives an equivalent reduced computation.

4 Computational method and numerical illustration

The rank differences in (1.13) can be computed by orthogonal elimination on the current obstruction space. We first give the exact algebraic algorithm and prove its equivalence to the subspace iteration. We then describe a QR/SVD realization with polynomial storage and arithmetic cost, avoiding the exponentially large matrices in (1.12).

4.1 Orthogonal elimination

Suppose $[Q_k \ V_k]$ is orthogonal, with $\text{im } V_k = \mathcal{V}_k$ and $\text{im } Q_k = \mathcal{V}_k^\perp$. Write $q_k = \dim \mathcal{V}_k^\perp$ and $d_k = n - q_k$, so that $Q_k \in \mathbb{R}^{n \times q_k}$ and $V_k \in \mathbb{R}^{n \times d_k}$. Initially, Q_0 has no columns and $V_0 = I_n$. By Lemma 1.1, $\mathcal{V}_{k+1} \subset \mathcal{V}_k$, so a candidate for the next space has the form $v = V_k b$, with $b \in \mathbb{R}^{d_k}$. Write the auxiliary vectors as $w_j = V_k a_j$, with $a_j \in \mathbb{R}^{d_k}$, and put $a = [a_1^\top \cdots a_m^\top]^\top$. Since a vector belongs to $\mathcal{V}_k$ precisely when $Q_k^\top$ annihilates it, the input and drift constraints become

$$B^\top V_k b + \sum_{j=1}^m D_j^\top V_k a_j = 0, \quad Q_k^\top A^\top V_k b + \sum_{j=1}^m Q_k^\top C_j^\top V_k a_j = 0.$$

The reduced block multiplying b is

$$L_k = \begin{pmatrix} B^\top V_k \\ Q_k^\top A^\top V_k \end{pmatrix} \in \mathbb{R}^{(r+q_k) \times d_k}, \quad (4.1)$$

and the block multiplying a is

$$M_k = \begin{pmatrix} D_1^\top V_k & \cdots & D_m^\top V_k \\ Q_k^\top C_1^\top V_k & \cdots & Q_k^\top C_m^\top V_k \end{pmatrix} \in \mathbb{R}^{(r+q_k) \times (md_k)}. \quad (4.2)$$

The constraints read

$$L_k b + M_k a = 0. \quad (4.3)$$

The coordinates of a already impose $w_j \in \mathcal{V}_k$ for every j , so these membership conditions require no additional equations.

To eliminate a from (4.3), let $U_{0,k}$ have orthonormal columns spanning $\ker M_k^\top$ and set

$$R_k = U_{0,k}^\top L_k. \quad (4.4)$$

Because $(\ker M_k^\top)^\perp = \text{im } M_k$, eliminating a from (4.3) gives

$$V_k b \in \mathcal{V}_{k+1} \iff L_k b \in \text{im } M_k \iff U_{0,k}^\top L_k b = 0 \iff b \in \ker R_k.$$

Thus $\mathcal{V}_{k+1} = V_k \ker R_k$. Let $\Delta q_k = \text{rank } R_k$, and choose an orthogonal matrix $O_k = [O_{1,k} \ O_{0,k}]$ whose first Δq_k columns span $(\ker R_k)^\perp$ and whose remaining columns span $\ker R_k$. The update is

$$Q_{k+1} = [Q_k \ V_k O_{1,k}], \quad V_{k+1} = V_k O_{0,k}. \quad (4.5)$$

The columns of $V_k O_{1,k}$ are the directions removed at this stage; they are appended to the orthogonal complement already computed.

This construction computes exactly the rank differences of the prescribed block pair.

Proposition 4.1. *In exact arithmetic, the updates in (4.1), (4.2), (4.4) and (4.5) preserve orthogonality of $[Q_k \ V_k]$ and satisfy $\text{im } V_k = \mathcal{V}_k$, $q_k = \delta_k$, and $\Delta q_k = \delta_{k+1} - \delta_k$. The iteration reaches a stationary space or the zero space after at most n stages, and returns the classification in Table 1 after checking $\text{rank } \mathcal{D}$.*

Proof. The elimination above gives $\text{im } V_{k+1} = V_k \ker R_k = \mathcal{V}_{k+1}$. Since V_k is an isometry on $\mathbb{R}^{d_k}$, the two blocks $V_k O_{1,k}$ and $V_k O_{0,k}$ are orthonormal and mutually orthogonal. Both are orthogonal to Q_k , so (4.5) preserves a complete orthogonal state basis and gives $q_{k+1} = q_k + \Delta q_k$. Starting from $q_0 = \delta_0 = 0$, (1.15) yields $q_k = \delta_k$ and the asserted increment identity.

If $\Delta q_k = 0$, then $\ker R_k = \mathbb{R}^{d_k}$ and $\mathcal{V}_{k+1} = \mathcal{V}_k$; by Lemma 1.1, all subsequent spaces coincide. Otherwise the dimension drops by at least one. Hence the algorithm terminates within n stages. The returned space is $\mathcal{V}_*$, and Theorem 1.2 gives the classification. $\square$

Algorithm 1 summarizes the exact-arithmetic computation.

Algorithm 1 Orthogonal elimination for the controllability criterion (exact arithmetic)

Require: $A, C_j \in \mathbb{R}^{n \times n}$, $B, D_j \in \mathbb{R}^{n \times r}$, $j = 1, \dots, m$.

```

1:  $Q \leftarrow []$ ,  $V \leftarrow I_n$ ; form  $\mathcal{D} = [D_1^\top \dots D_m^\top]^\top$ .
2: for  $k = 0, \dots, n-1$  do
3:   Form  $L$  and  $M$  from  $Q, V$  by (4.1) and (4.2).
4:   Reduce  $M$  by left orthogonal transformations and apply the same transformations to  $L$ .
5:   Let  $R$  consist of the rows of the transformed  $L$  corresponding to the zero rows of the transformed  $M$ .
6:   Find  $\Delta q = \text{rank } R$  and an orthogonal matrix  $O = [O_1 \ O_0]$ , with  $\text{im } O_0 = \ker R$ .
7:   if  $\Delta q = 0$  then
8:     return nonzero subspace  $\text{im } V$ ; all four controllability properties fail.
9:   end if
10:   $Q \leftarrow [Q \ VO_1]$ ,  $V \leftarrow VO_0$  (using the old  $V$  on both right sides).
11:  if  $V$  has no columns then
12:    return AC, ANC and ENC hold; EC holds if and only if  $\text{rank } \mathcal{D} = mn$ .
13:  end if
14: end for
```

4.2 Numerical realization and complexity

The elimination uses Householder QR and SVD [2]. At one stage, put $p = r + q$, $d = n - q$, $h = md$ and $s = \text{rank } M$, so $M \in \mathbb{R}^{p \times h}$ and $L \in \mathbb{R}^{p \times d}$. When $p \geq h$, apply Householder QR to $[M \ L]$, then an SVD to the leading $h \times h$ triangular block of the transformed M . Apply the same left transformations to L and retain all $p - s$ rows corresponding to zero rows of M . These form R . Reflectors avoid storing a full $p \times p$ factor. When $p < h$, a full left singular-vector matrix of M supplies these rows directly; its full right factor is unnecessary.

An SVD of R gives the decomposition in (4.5). Its full $d \times d$ right factor is required when R is wide, since a thin SVD omits part of the nullspace. If R has no rows, its kernel is $\mathbb{R}^d$ and the iteration stops. After updating, Householder QR may reorthogonalize $[Q \ V]$ in its existing order.

The use of the residual itself is important for numerical rank decisions. Indeed,

$$R_k^\top R_k = L_k^\top (I - \Pi_{\text{im } M_k}) L_k.$$

This Gram matrix has the correct kernel, but its nonzero eigenvalues are the squares of the nonzero singular values of R_k . We therefore compute the singular values of R_k directly. We also avoid forming the projected residual by the subtraction $L_k - U_{1,k}(U_{1,k}^\top L_k)$, where $U_{1,k}$ spans $\text{im } M_k$ orthonormally: cancellation can obscure a small residual. Applying the orthogonal transformations to L_k and extracting the required rows gives (4.4) without this subtraction.

In floating-point arithmetic, ranks must be decided relative to a stated scale and tolerance. For each matrix X whose rank is required, one may count singular values above $\tau_{\text{abs}} + \tau_{\text{rel}} \|X\|$, with tolerances chosen for the data scale and precision. Singular values near this threshold require a sensitivity check; the orthogonal implementation does not remove the nonrobustness in Example 3.1.

The coefficient matrices, the current orthogonal state basis, and the work arrays require $O(mn(n+r))$ storage. Indeed, $p \leq r + n$ and $h \leq mn$; the tall branch does not store a full $p \times p$ factor, while in the wide branch $p^2 < ph$. The factorization of M costs at most $O(ph \min(p, h)) \leq O(ph^2)$ arithmetic operations. Including matrix products, the decomposition of R , and reorthogonalization, each stage costs at most $O(m^2 n^2 (n+r))$. At most n stages and the final singular-value computation for $\mathcal{D}$ therefore cost $O(m^2 n^3 (n+r))$ in dense real arithmetic. In contrast, at $n = 14$, $r = 2$, and $m = 1$, the prescribed matrix N_n alone would require about 56 GiB of dense storage at eight bytes per entry, by the dimensions in (1.12).

4.3 Input configurations and control energy

We compare the energy required to reduce the terminal error under three input configurations of the same two-dimensional system with two Brownian motions. Take $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $C_1 = \begin{pmatrix} 1/2 & 0 \\ 1 & 1/2 \end{pmatrix}$ and $C_2 = \text{diag}(1/3, -1/3)$. The five candidate input columns are given by $B = e_2(1, 1, 0, 0, 0)$, $D_1 = [0 \ e_2 \ e_1 \ e_2 \ 0]$ and $D_2 = [0 \ e_1 \ 0 \ 0 \ e_2]$,

Table 2: Input classification by Corollary 1.5.

Case	(δ_1, δ_2)	rank $\mathcal{D}$	Properties
I	(0, 0)	1	None
II	(1, 2)	1	AC, ANC, ENC
III	(1, 2)	4	AC, ANC, ENC, EC

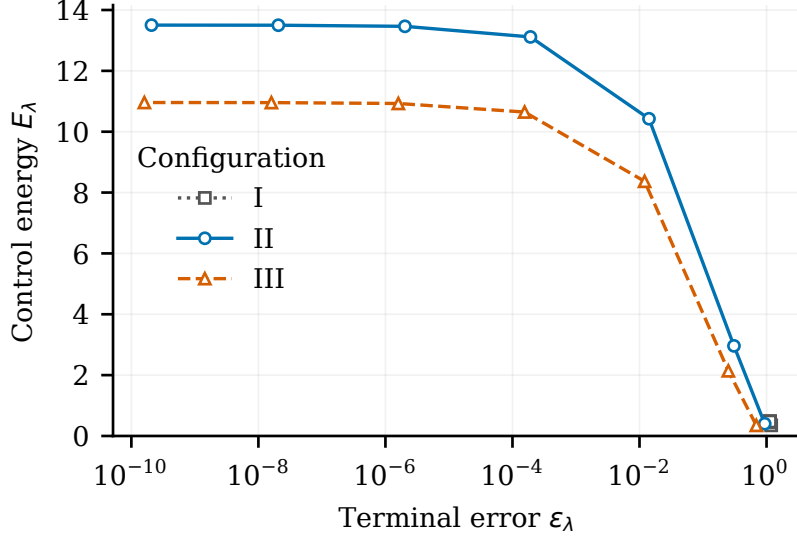


Figure 1: Terminal error and expected control energy for $\lambda = 10^k$, $k = 0, \dots, 6$, computed from the Riccati and second-moment equations. Markers show all seven penalties in each configuration.

where e_1, e_2 are the coordinate vectors and 0 denotes a zero column. Configurations I, II and III use columns $\{2\}$, $\{1, 2\}$ and $\{1, 2, 3, 4, 5\}$, respectively, in all three input matrices.

Orthogonal elimination and exact rational rank calculations give Table 2. Adding the drift-only input in II removes the geometric obstruction, while the additional diffusion inputs in III make $\mathcal{D}$ surjective. Thus II permits exact null control, and III also permits exact control to arbitrary random targets.

For $T = 1$ and $y_0 = e_1$, we solve

$$\min_{u \in \mathcal{U}_1} \left\{ \mathbb{E} \int_0^1 |u(t)|^2 dt + \lambda \mathbb{E} |y(1)|^2 \right\},$$

with $\lambda = 10^k$, $k = 0, \dots, 6$. The Riccati method in subsection 2.2, with drift A , zero running state weight and terminal weight λI_2 , gives the optimal feedback $u_\lambda = K_\lambda y_\lambda$. We compute $M_\lambda(t) = \mathbb{E}[y_\lambda(t)y_\lambda(t)^\top]$ from

$$\begin{cases} \dot{M}_\lambda = F_\lambda M_\lambda + M_\lambda F_\lambda^\top + \sum_{j=1}^2 G_{\lambda,j} M_\lambda G_{\lambda,j}^\top, \\ M_\lambda(0) = e_1 e_1^\top, \end{cases}$$

where $F_\lambda = A + BK_\lambda$ and $G_{\lambda,j} = C_j + D_j K_\lambda$. The terminal error is $\epsilon_\lambda = \text{tr } M_\lambda(1)$ and the control energy is

$$E_\lambda = \int_0^1 \text{tr}(K_\lambda(t) M_\lambda(t) K_\lambda(t)^\top) dt.$$

Figure 1 shows a persistent terminal error in I, whereas II and III attain decreasing errors with energies that level off. In II, increasing λ from 10^2 to 10^6 reduces the error from 1.4105×10^{-2} to 2.0745×10^{-10} , while the energy increases from 10.426 to 13.503. At $\lambda = 10^6$, III gives an error of 1.6017×10^{-10} with energy

10.959, while I retains an error of about 1.1293. The behavior in II and III agrees with the bounded-energy approximation provided by ENC. Their different EC classifications are determined by the joint diffusion rank in Table 2.

5 Conclusions

We have obtained a unified finite algebraic classification of four controllability properties for constant-coefficient linear stochastic systems with multidimensional Brownian noise and arbitrary joint diffusion rank. A common geometric obstruction characterizes approximate controllability, approximate null controllability and exact null controllability, while exact controllability additionally requires surjective diffusion control. The feedback construction provides finite-energy null control at any prescribed positive time, and orthogonal elimination makes the rank tests computable without forming exponentially large block matrices. The numerical example relates this classification to the terminal-error and energy behavior of penalized LQ controls under successive additions of input channels. Quantifying the effects of bounded gains and finite switching resolution is a natural direction for further work.

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